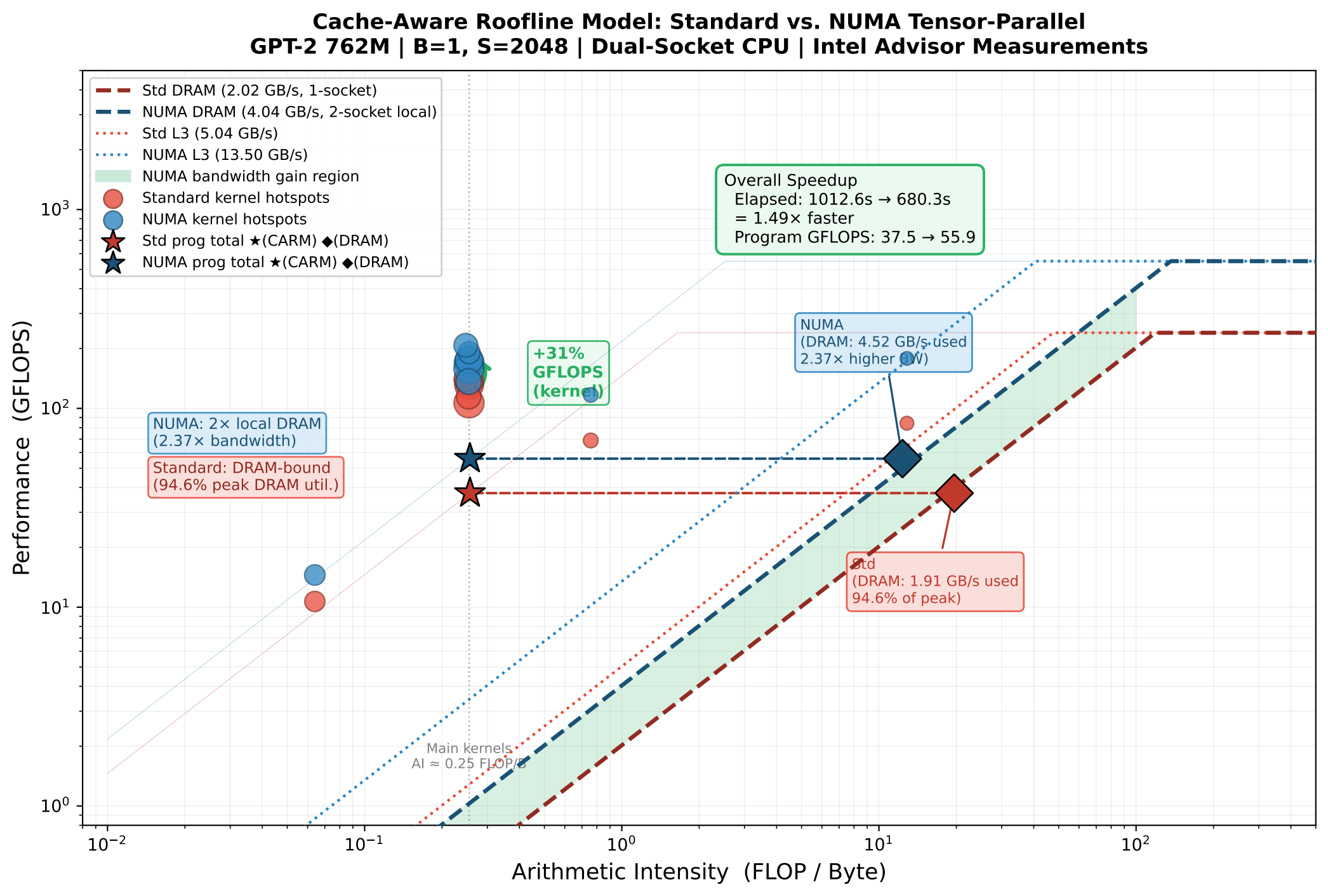

Cache-Aware Roofline Model: Standard vs. NUMA Tensor-Parallel
GPT-2 762M | B=1, S=2048 | Dual-Socket CPU | Intel Advisor Measurements
Std DRAM (2.02 GB/s, 1-socket)
NUMA DRAM (4.04 GB/s, 2-socket local)
Std L3 (5.04 GB/s)
NUMA L3 (13.50 GB/s)
NUMA bandwidth gain region
Standard kernel hotspots
NUMA kernel hotspots
Std prog total ★(CARM) ◆(DRAM)
NUMA prog total ★(CARM) ◆(DRAM)
Overall Speedup
Elapsed: 1012.6s → 680.3s
= 1.49× faster
Program GFLOPS: 37.5 → 55.9
+31%
GFLOPS
(kernel)
NUMA
(DRAM: 4.52 GB/s used
2.37× higher BW)
NUMA: 2× local DRAM
(2.37× bandwidth)
Standard: DRAM-bound
(94.6% peak DRAM util.)
Std
(DRAM: 1.91 GB/s used
94.6% of peak)
Main kernels
AI ≈ 0.25 FLOP/B
Performance (GFLOPS)
Arithmetic Intensity (FLOP / Byte)